

PROJECT PROPOSAL

Can Vulnerability-Guided KL Anchors Reduce Catastrophic Forgetting in Parameter-Efficient Fine-Tuning?

A controlled study of pilot-based anchor selection during LoRA adaptation

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Project Summary

We propose a controlled study of whether a small set of carefully selected, unlabeled prompts can reduce general-capability degradation during Low-Rank Adaptation (LoRA). We will begin with Qwen2-1.5B-Instruct, improve its mathematical word-problem performance by fine-tuning on GSM8K, and measure changes on three unrelated retention benchmarks: BoolQ, HellaSwag, and ARC-Easy. LoRA freezes the pretrained backbone and learns small low-rank updates, but those shared updates still affect predictions for every input and can therefore alter unrelated behavior [1, 2].

A frozen copy of the original model will act as a teacher. A temporary pilot LoRA will reveal which general prompts experience the largest teacher-to-pilot change in token probabilities. We will discard that pilot, restart every main run from the same base checkpoint, and use the vulnerable prompts as KL anchors during training. The study asks whether this vulnerability-guided use of a fixed protection budget preserves more capability than random or semantic-diversity selection, without obtaining an advantage merely by learning less of the target task.

1 Goals and Objectives

Scientific Question

Under equal target-training, anchor-token, and teacher-compute budgets, can a short pilot LoRA identify unlabeled prompts whose use as KL anchors reduces catastrophic forgetting more effectively than random or diversity-only anchors at matched GSM8K performance?

Objectives

- Establish a reproducible task-only LoRA baseline and verify that mathematical adaptation produces measurable target improvement and general-capability change.
- Develop a pilot-based vulnerability score using token-level KL divergence between the frozen teacher and a briefly adapted student.
- Compare random, semantic-diversity, highest-damage, and diverse-damage anchor selection with the same number of anchor response tokens and comparable training cost.
- Distinguish aggregate score changes, item-level forgetting, output-format failures, and prompt-recoverable errors.

Hypotheses

- H1 Damage-KL and Diverse-Damage-KL will produce lower average retention loss than Random-KL at matched GSM8K accuracy.
- H2 Diverse-Damage-KL will outperform simple top-damage selection when the highest-scoring prompts contain many near-duplicates.
- H3 Any genuine retention advantage will remain visible under forced-choice likelihood scoring and alternative prompt templates, rather than appearing only through easier answer formatting.

Research Originality and Scope

LoRA, knowledge distillation, KL regularization, replay, and diversity sampling are established techniques [1, 3, 4]. Prior work has shown that LoRA can forget less partly because it learns less [2], and recent work predicts forgotten examples for selective labeled replay [5]. Our attempted contribution is narrower: pilot-measured behavioral vulnerability will guide the selection of unlabeled KL anchors, and the selectors will be compared under a fixed budget, matched target learning, and recovery-aware evaluation. We will present this as a new controlled combination and empirical research question, not as the invention of KL anchoring or an absolute first claim. The initial study is limited to one model and a mathematics adaptation; a second target dataset or domain will be attempted only after the primary matrix is complete.

2 Tasks

Primary Adaptation Task

The student model will be fine-tuned with supervised LoRA on GSM8K, a public collection of about 8,500 grade-school mathematical word problems with natural-language solutions [9]. The official training split will be divided into training and development subsets, while the official test split will remain untouched until final evaluation. The target metric will be exact match on the extracted final numerical answer.

Example	Content
Input	A shop has 18 notebooks and sells 7. How many notebooks remain?
Expected output	The shop has $18 - 7 = 11$ notebooks remaining. Final answer: 11.

Retention Tasks

We will evaluate abilities that are unrelated to GSM8K and that the untouched Qwen2 model can perform above chance. BoolQ measures yes-or-no reading comprehension [10]. HellaSwag measures commonsense continuation selection [11]. ARC-Easy measures multiple-choice grade-school science reasoning [12]. These tasks cover different response structures and reduce the risk that one benchmark-specific prompt format determines the conclusion.

Anchor Selection Task

Approximately 2,000 general instructions will form an unlabeled candidate pool. For each prompt, the frozen teacher will generate one short deterministic continuation. The teacher and pilot student will then score exactly the same prompt-continuation tokens. The average teacher-to-pilot KL divergence over response positions will be the prompt's vulnerability score. A high score means the pilot changed its output distribution strongly on that input; it does not prove that a particular parameter or fact was erased.

Semantic diversity refers to diversity of meaning rather than wording. We will embed each prompt as a numerical sentence representation and cluster prompts by cosine similarity [13]. Diversity-KL will sample across clusters. Diverse-Damage-KL will favor high-vulnerability prompts while limiting repeated prompts from the same semantic cluster.

3 Data

Dataset	Role	Use and controls
GSM8K	Target adaptation	Train LoRA on the official training data; create a fixed development split; reserve the official test split for final numerical exact match.
Databricks Dolly 15k	Anchor candidates	Use about 2,000 instruction fields without gold responses. Remove empty, unsafe, overly long, mathematics-like, and duplicate prompts [14].
BoolQ	Retention	Use a held-out slice of the training split for protocol checks and the official validation split for final accuracy.
HellaSwag	Retention	Use the official validation split for final length-normalized option likelihood and accuracy.
ARC-Easy	Retention	Use the official validation split for final answer-option likelihood and accuracy.

All target examples, anchor candidates, retention-development items, and final evaluation items will remain disjoint. We will record dataset versions and exact indices. Exact matching, normalized-text matching, and sentence-embedding similarity will be used to remove duplicates and suspicious overlap. Final benchmark questions and their answer choices will never be used as anchors or training examples.

4 Models and Methods

Model Configuration

The base checkpoint will be Qwen/Qwen2-1.5B-Instruct, an open instruction-tuned decoder model from the Qwen2 family [8]. The same checkpoint, tokenizer, chat template, maximum sequence length, and LoRA target modules will be used for every condition. We will begin with rank 16 LoRA and freeze the final configuration after a task-only development run. BF16 LoRA will be preferred; four-bit loading may be used only as a documented hardware fallback and will then be applied identically to all trained conditions.

Teacher and Student

The teacher is an unchanged copy of Qwen2-1.5B-Instruct. The student shares the frozen backbone but activates trainable LoRA matrices. On target batches, the student learns the GSM8K response. On anchor batches, it is penalized when its next-token distribution differs from the teacher's distribution. The total objective therefore combines task loss with a weighted anchor KL loss. The KL weight will be chosen from a small development sweep and then fixed before final evaluation.

An anchor does not store an update for only one prompt. Its loss produces gradients through the same LoRA matrices used by all inputs, so it can stabilize related behaviors and can also create side effects. This shared influence is why held-out retention evaluation and equal-budget comparisons are necessary.

Selection stage

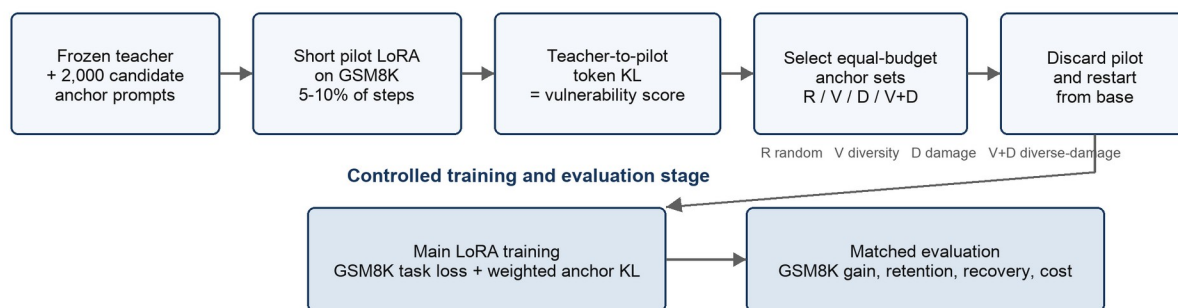


Figure 1 Pilot-guided selection followed by a fresh, controlled main training stage

Experimental Procedure

- Step 1 Establish the reference by evaluating the untouched model and caching deterministic teacher continuations for every candidate anchor.
- Step 2 Train a temporary pilot LoRA for approximately 5-10 percent of the planned GSM8K steps, using fixed data indices and a recorded seed.
- Step 3 Calculate response-token KL drift for every candidate and construct four equal-budget anchor sets: random, diversity, damage, and diverse-damage.
- Step 4 Discard the pilot. Start every main condition from the identical base checkpoint and train with the same target examples, target tokens, optimizer schedule, and LoRA configuration.
- Step 5 Match anchor response tokens, teacher forward passes, and total optimizer work as closely as practical. Save regular checkpoints for target-performance matching.
- Step 6 Evaluate GSM8K learning, general retention, item transitions, recoverability, and cost using frozen prompts and scripts.

Downloaded and Implemented Components

We will download the Qwen2 checkpoint, public datasets, Hugging Face Transformers and Datasets, PEFT, a sentence-embedding model, and the Language Model Evaluation Harness. We will implement the pilot protocol, fixed-continuation teacher cache, token-masked KL loss, vulnerability and diversity selectors, equal-budget controller, target-matched checkpoint comparison, recovery battery, item-level transition analysis, and reproducible experiment configuration. The proposed originality lies in this selection and evaluation protocol.

5 Baselines

Condition	Anchor selection	Purpose
Untouched base	None	Reference for initial capability and zero adaptation.
Task-only LoRA	None	Measures ordinary target learning and forgetting without protection.
Random-KL	Uniform random	Tests whether any small general anchor set is sufficient.
Diversity-KL	Semantic clusters	Tests whether broad coverage matters without vulnerability information.
Damage-KL	Highest pilot KL	Tests direct vulnerability-guided protection.
Diverse-Damage-KL	High KL across clusters	Tests vulnerability guidance with redundancy control; proposed primary method.

The primary comparison will be Diverse-Damage-KL versus Random-KL at a middle budget, initially 128 prompts. Damage-KL and Diversity-KL comparisons will be secondary. In a different Llama-2-7B and MetaMathQA setup, Biderman et al. reported GSM8K accuracy of 62.2 percent for LoRA versus 64.2 percent for full fine-tuning, with retention averages of 0.63 and 0.57 respectively [2]. These are contextual scores, not direct baselines; our conclusions will use within-Qwen2 comparisons under recorded evaluation settings [15].

6 Evaluation

Quantitative Metrics

Question	Metric	Interpretation
Did the model learn mathematics?	GSM8K numerical exact match and target negative log-likelihood	Higher exact match and lower loss indicate stronger target adaptation.
Did general capability decline?	Accuracy on BoolQ, HellaSwag, and ARC-Easy; base minus adapted score	Report each task and an equal-weight macro average, not only one aggregate.
Which individual items changed?	Correct-to-incorrect and incorrect-to-correct transitions	Separates forgetting from positive backward transfer [7].
Is the comparison fair?	Matched GSM8K checkpoints and target-gain versus forgetting frontier	Prevents a weak learner from appearing to retain more simply because it changed less [2].
Was the protection efficient?	Retention gain per 100,000 anchor tokens and per GPU-hour	Controls for additional data and teacher computation.
Is the loss recoverable?	Alternative-template and fixed few-shot recovery rates	Separates prompt-access or format failures from residual behavioral loss [6].

Target Matching and Recovery Controls

We will compare checkpoints whose GSM8K validation accuracies are within approximately one percentage point. If no such pair exists, we will present the full target-gain versus forgetting frontier. Multiple-choice retention will primarily use answer-option likelihood. Fixed alternative templates and development-split few-shot demonstrations will test whether failures are prompt-recoverable. Remaining errors will be called residual behavioral loss, not proven erasure of internal knowledge.

Statistics and Qualitative Analysis

Central conditions will use at least two seeds. We will report seed-level results and paired bootstrap 95 percent confidence intervals. A stratified error sample will distinguish reasoning or factual mistakes from prompt misinterpretation, answer-format failures, refusal, and excessive generation. We will also inspect which semantic clusters dominate the high-vulnerability anchors.

7 Feasibility and Project Plan

Minimum Viable Study

The essential study contains five trained LoRA conditions - task-only, Random-KL, Diversity-KL, Damage-KL, and Diverse-Damage-KL - with two seeds, plus the frozen base reference and a short pilot. This produces ten central training runs on a 1.5B-parameter model. The primary anchor budget will be fixed before the final runs. Anchor-budget sweeps, additional pilot seeds, a second model, and a second adaptation dataset or domain will begin only after this matrix and its recovery analysis are complete.

Risks and Fallbacks

- **Little measurable forgetting:** Increase target exposure only within a documented range and inspect checkpoint trajectories. If the effect remains too small, extend the adaptation data with a public mathematics-instruction corpus while keeping GSM8K as the fixed target test. A second non-mathematical target is a stretch goal, not a required result.
- **KL computation is expensive:** Cache teacher continuations and either full logits or a documented top-k approximation. Apply the same approximation to every KL condition.
- **Pilot rankings are unstable:** Measure rank correlation across pilot checkpoints or seeds. If instability persists, report it as a finding and avoid claiming that vulnerability is predictable.
- **KL protects retention by blocking learning:** Tune the KL weight only on development data, compare matched-GSM8K checkpoints, and report the full learning-retention frontier.
- **Time or hardware is limited:** Prioritize task-only, Random-KL, Damage-KL, and Diverse-Damage-KL with two seeds; omit stretch datasets and broad hyperparameter sweeps before reducing evaluation rigor.

Expected Outputs

The project will produce a reproducible training and evaluation pipeline, fixed data manifests, vulnerability scores and selected anchor sets, learning-retention Pareto plots, item-transition and recovery analyses, compute-normalized comparisons, and a report that includes positive, null, or negative findings. A null result with tight uncertainty and documented failure analysis will still answer the scientific question.

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Reproducibility Note

Published scores will be used only when the model, dataset split, prompt, and scoring protocol are sufficiently comparable. The final report will identify preprints separately from peer-reviewed publications and will archive the exact versions of code and data used in the study.